AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 9 · TEACHER GUIDE

# Prompt Evaluation

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

## Teach selection as a rule applied to complete evidence.

Supply all source/prompt/output cards, preserve failed cases and assess the fresh five-case record.

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**Keep one habit:** Ask which cases differ, which gate was declared and what remains unknown after selection.

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MAIN LESSON · SECTION 1

## Prepare this connected week

> Use one main plan and keep practice separate from the independent first attempt.

### Learning goal and precision

Primary target: Explain the **mechanism** and terminology of controlled prompt experiments

Two 45-minute sessions. A prompt experiment compares complete specifications using the same task cases. Define success before inspecting **output**s; change one intended instruction and preserve other settings.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

By the end, I can evaluate prompts with **paired case**s and a predeclared mandatory criterion.

## What to prepare

- Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.
- Case exposure check: this source example appears in ai-2/28, ai-3/7, ai-3/9. Reuse a familiar case for recall or modelling; verify that the case used to assess transfer has not already received feedback.
- Lesson and workbook; paper and pencil
- Calculator and optional local Python 3 for arithmetic; no live API or private data
- Prepare the fresh retry separately and withhold its solution until after collection

## Main teaching plan

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

## Opening retrieval and readiness

Without opening the old lesson, explain one key idea from Structured **Output** and Schemas. Give an example and a mistake that the idea helps you avoid.

Look for: Separate JSON parsing, schema validation and source support checks

Reference example, not the only acceptable wording: Teacher explanation: Parsing establishes grammatical JSON; schema checks enforce fields/types/conditional rules; semantic checks match the claims to sources and domain meaning.

None alone establishes all three.

Without opening the old lesson, explain one key idea from Examples and **Rubric**s. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish a format example from approved factual evidence

Reference example, not the only acceptable wording: A format example demonstrates structure; approved N supplies the frog fact. The original mammal answer fails mandatory support even with3/4points, while the answer-only repair still lacks the Question field. Check each required criterion separately.

Without opening the old lesson, explain one key idea from Tokens and Context. Give an example and a mistake that the idea helps you avoid.

Look for: Account for fixed text, **evidence** and reserved **output** under a toy budget

Reference example, not the only acceptable wording: Teacher explanation: 100-20-10-20 = 50 supplied token units.

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

▼ **Earlier wording, alternatives and picture transcripts**

**Teach the learning cycle**

**Retrieval answer guidance**

**Structured Output and Schemas**

**Examples and Rubrics**

**Tokens and Context**

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

**What the learner must demonstrate**

On four shared tasks, A **score**s 1,1,0,1 and B **score**s 1,0,1,1. Their totals tie. What should you inspect before choosing? Define a final evaluation step.

**Concept to explain**

Prompt claims require a fixed **test set** and repeatable scoring.

| Term | Meaning |
| --- | --- |
| A/B test | A comparison of two versions using a planned assignment of comparable cases and the same outcome measure. |
| test set | Data reserved for evaluating a selected system; final test data must not influence its development. |
| score | A numerical value assigned using a specified rule; it is not automatically a probability. |
| variance | A measure of spread around a mean; informally, variation across repeated results. Specify which meaning is intended. |

### Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

### Preparation for the connected explanation

### Readiness answer

### End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

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MAIN LESSON · SECTION 2

## Model paired comparison and predeclared selection

> Keep P/Q review and A/B refusal-gate data separately labelled.

## State fixed criteria and evidence limits

A prompt experiment compares complete specifications using the same task cases. Define success before inspecting **output**s; change one intended instruction and preserve other settings.

Evaluate both prompts on **paired case**s with fixed model/version, source records, **output** contract and scoring **rubric**. Use development results for selection, then reserve new final cases.

**Score** each case rather than counting attractive sentences.

## Trace the separate P/Q review

| Separate six-case P/Q review | P | Q |
|---|---|---|
| Passing IDs | 1,2,3,4 | 1,2,3,5,6 |
| Missing-approval slice4,6 | 1/2 | 1/2 |
| Overall | 4/6 | 5/6 |

On six development cases P passes cases 1, 2, 3, 4; Q passes 1, 2, 3, 5, 6. Missing-approval cases are 4 and 6. Compare totals and condition results.

P passes 4/6 and Q 5/6. P passes one of two missing-approval cases, Q also one of two. Q improves the aggregate while leaving this condition unchanged.

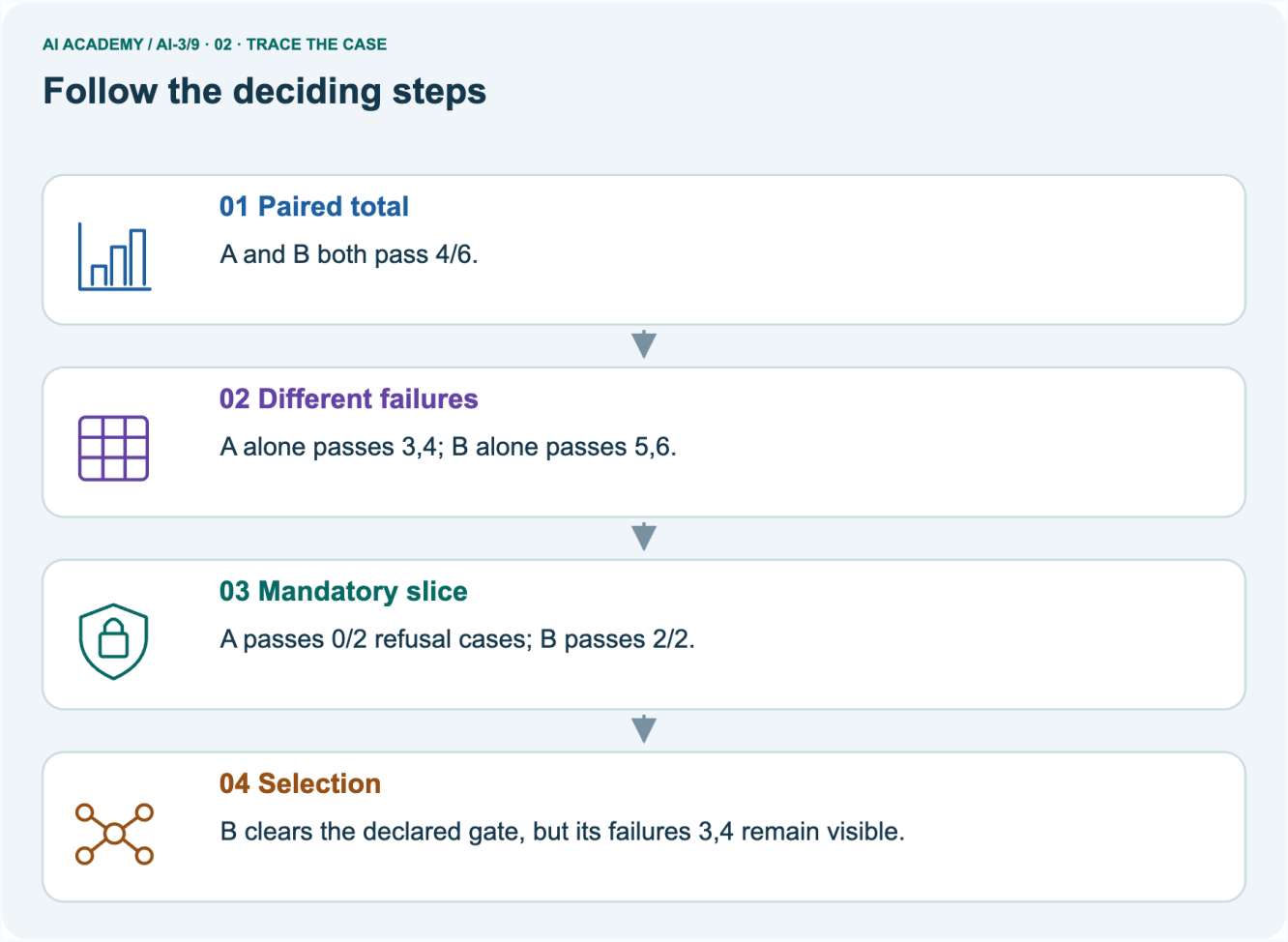
## Read the pictured A/B cases and gate

A fictional Study Buddy compares two prompts across six development questions.

A passes cases 1,2,3,4. B passes 1,2,5,6. Cases 5 and 6 require refusing unsupported claims.

Mandatory selection gate: pass both refusal cases. All cases use fixed sources and one common **rubric**.

## Trace totals, differences and mandatory slice




Follow the labelled picture with the matching explanation. Open the vector diagram

### Read the picture one step at a time

| STEP | WHAT HAPPENS AND WHY |
| --- | --- |
| 1. Paired total | A and B both pass 4/6. |
| 2. Different failures | A alone passes 3,4; B alone passes 5,6. |
| 3. Mandatory slice | A passes 0/2 refusal cases; B passes 2/2. |
| 4. Selection | B clears the declared gate, but its failures 3,4 remain visible. |

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Equal totals do not imply equal behavior. The predeclared gate expresses an important requirement instead of choosing a criterion after seeing the result.

Selecting B is justified under that rule, not by claiming it is best on every question. After development, freeze the chosen prompt and test new questions, including both ordinary and unsupported cases.

## Predict removing the gate

Change: Remove the gate and retain total passes as the only criterion.

**Prediction**: The record is a tie; another predeclared tie policy or further development **evidence** is needed.

Reason: Both prompts have four passes on six questions.

## State small-invented-suite limits

Six invented development outcomes do not measure real model performance or prove a gain in student understanding.

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

**Input** → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. Prompt A **score**s 20/30 across six cases.

2. Prompt B **score**s 25/30 on the same cases and **rubric**.

3. B improves **evidence** but loses one clarity point on long passages.

4. Repeat runs and inspect per-case failures before choosing.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

### Concrete teaching case

On five fixed questions, A passes four and B passes three; B looks more attractive.

What conclusion is supported?

### Exact explanation to model

A has 4/5 versus B's 3/5 on this set. Appearance is separate. Broaden cases and repeat stochastic trials before generalising.

### Explain the added worked example

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

### Key terms

| Term | Meaning |
| --- | --- |
| Paired case | The same case evaluated under each compared procedure |
| Rubric | Prespecified criteria for judging an output |
| Regression | A previously passing condition that fails after a change |

### Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

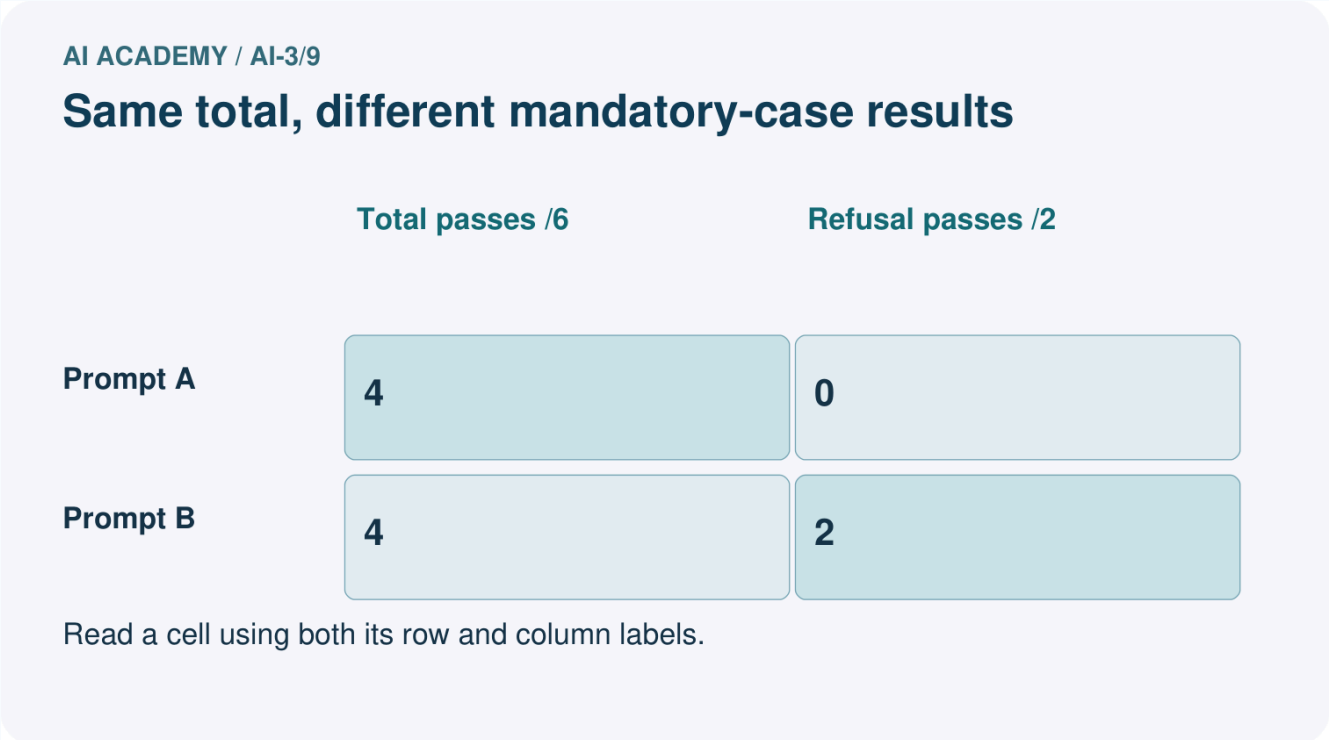
The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

| Thinking move | Teacher prompt |
| --- | --- |
| Plan | What is given? What is the question? Which procedure fits these facts? |
| Do | Point to each input and intermediate step in the worked trace. Name the rule that allows the next step. |
| Check | Does the output answer the question? Which assumption or limitation prevents a larger claim? |

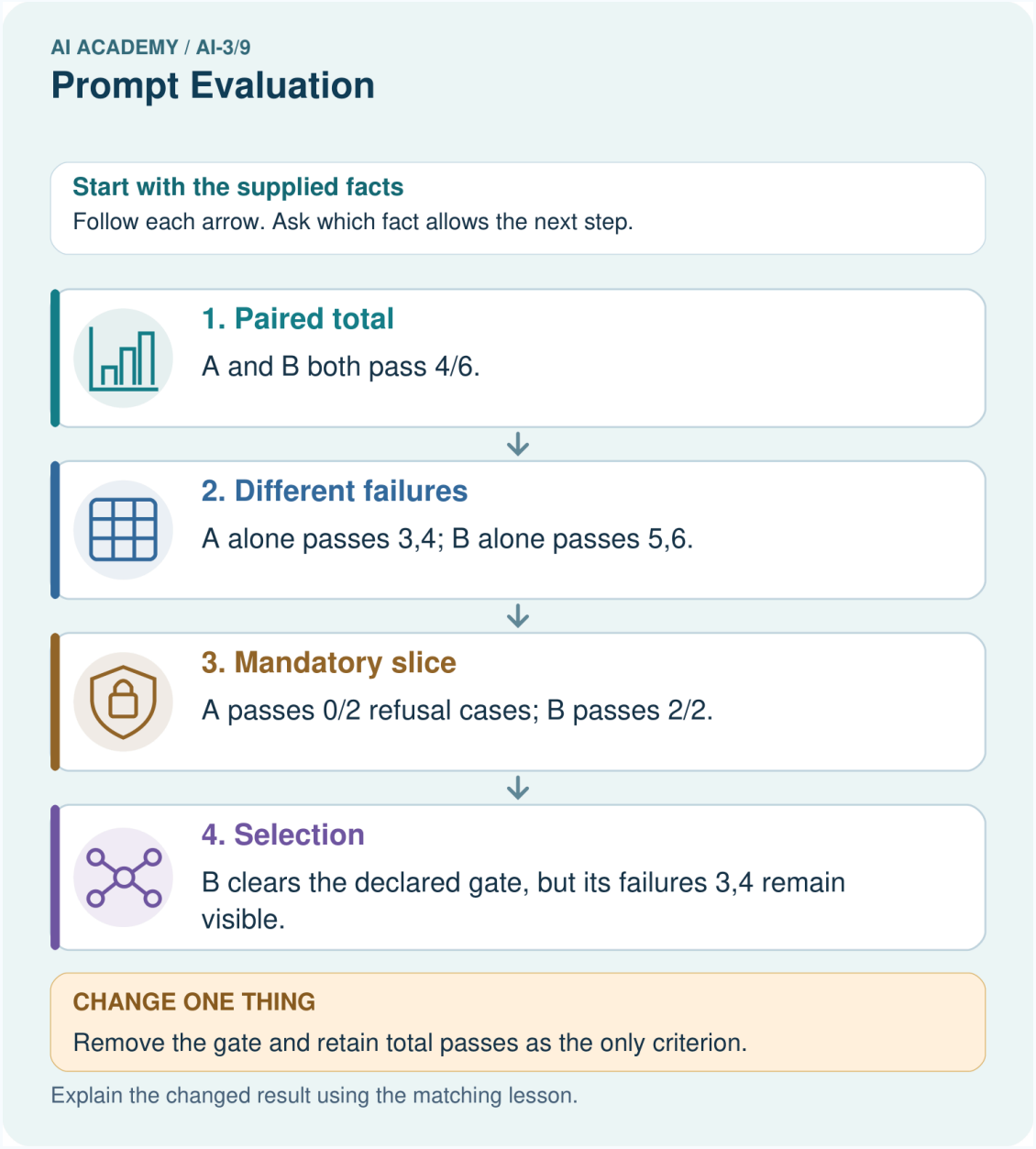
Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

### A pictorial worked case: Prompt Evaluation

### The facts we will use




Follow the labelled picture with the matching explanation. Open the vector diagram




Follow the labelled picture with the matching explanation. Open the vector diagram

## Why this result follows

## What this example does not establish

## Change one thing in the pictured case

## Ask while showing the picture

MAIN LESSON · SECTION 3

# Guide the explanation, then remove help

> Collect each learner’s reasoning before showing feedback.

### Main completion prompt

Completion task: On six development cases P passes cases 1, 2, 3, 4; Q passes 1, 2, 3, 5, 6. Missing-approval cases are 4 and 6. Compare totals and condition results.

Show the intermediate steps and give the conclusion with its scope.

## Expected reasoning and fading help

Teacher check: Teacher explanation: P passes 4/6 and Q 5/6. P passes one of two missing-approval cases, Q also one of two. Q improves the aggregate while leaving this condition unchanged.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

## Individual misconception check

Use the worked case for Controlled prompt experiments. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Prompt selection on development cases cannot be reported as independent final performance.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

## Pictorial question and changed-case reasoning

Why is B's selection defensible here without calling it universally superior?

It alone passes both mandatory refusal cases. A is better on cases 3 and 4, which must still be reported.

Changed-case reasoning: The record is a tie; another predeclared tie policy or further development **evidence** is needed. Both prompts have four passes on six questions.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

## Keep each learner explaining

- Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.
- Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.
- In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

### ▼ Earlier wording, alternatives and picture transcripts

Demonstrate and guide practice

#### Board sequence

Write the given facts: On five fixed questions, A passes four and B passes three; B looks more attractive.

Write the question to explain: What conclusion is supported?

Reveal after **prediction**s: A has 4/5 versus B's 3/5 on this set. Appearance is separate. Broaden cases and repeat stochastic trials before generalising.

Use the earlier changed question as optional later practice. Reserve Compare new paired flags independently for the first independent attempt before hints or feedback.

### Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

### Model these guided questions

Guided: P passes 4/6 and Q 5/6. P passes one of two missing-approval cases, Q also one of two. Q improves the aggregate while leaving this condition unchanged.

Qualification: Prompt selection on development cases cannot be reported as independent final performance.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

### Case-specific teaching sequence

| Stage | Teacher action |
|---|---|
| Session 1: 0–7 minutes | Retrieve the prerequisite; identify the input, procedure and evidence source. |
| 7–18 minutes | Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation. |
| 18–32 minutes | Trace the declared procedure and worked case. Complete W2, recording intermediate steps. |
| 32–45 minutes | Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations. |
| Session 2: 0–8 minutes | Retrieve the worked reasoning and collect the guided response before discussing it. |
| 8–20 minutes | Compare a proposed boundary variant with the original case; identify what remains fixed. |
| 20–35 minutes | Collect W4 independently. Require the procedure trace and the bounded conclusion. |

| Stage | Teacher action |
| --- | --- |
| 35–45 minutes | Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts. |

### Reduce help deliberately

### Individual reasoning check

### Pictorial practice question

### Reasoned teacher answer

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MAIN LESSON · SECTION 4

# Run the supplied investigation

> Keep expected results, observed paper traces and remaining limits distinguishable.

## Supplied materials and answer guidance

Main A/B cards:1,2 passboth;3Apass/Bwrongterm;4Apass/BmissingMeaning;5,6Aunsupported/Bnot supplied. Both4/6, shared1,2;Aonly3,4;Bonly5,6. Refusal gateA0/2,B2/2 selectsB under that rule;B failures3,4remain. Withoutgate totals tie. Separate P/Q review4/6versus5/6 with slice1/2both. No real model performance, blind reliability or causal effect measured. Reserve the selected independent five-case comparison.

## Run and record

1. Predict → Test → Record → Explain
2. Keep expected and actual results separate. Surprises are useful data when recorded honestly.

3. Audit the six complete fictional A/B output cards using approved N and the predeclared refusal gate. **Score** each criterion, compare paired outcomes and preserve remaining failures. Do not label the supplied records actual model runs, blind ratings after disclosure or measured causal gains.

4. What did the hardest test teach you that the normal test did not?

## Keep an inspectable record

| Record | What to capture |
| --- | --- |
| Input or evidence | Exact data, instruction or claim |
| Expected result | Answer or safe fallback declared before the run |
| Observed result | Actual result, including errors |
| Explanation | Rule, calculation or evidence explaining the outcome |
| Next test | One justified change and a new prediction |

## Diagnose the first misunderstanding

The worked result establishes every future case: Prompt selection on development cases cannot be reported as independent final performance.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

## Choose support from the response

| What the response shows | Next teaching action |
|---|---|
| Missing or misread facts | Read the case aloud, mark supplied versus unknown facts, and let the learner restate it. |
| Wrong procedure | Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation. |
| Arithmetic or implementation error | Trace intermediate values or lines; preserve the first mismatch and retry that step. |
| Correct result without a reason | Ask for a trace and a changed case before marking independent understanding. |
| An unsupported wider claim | Ask which supplied evidence supports the claim and which further observation would be needed. |

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

### ▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

### Make the lab runnable

Use the supplied investigation for shared practice. Keep predictions and observed paper-trace results separate. Reserve Compare new paired flags independently for the individual first attempt before feedback; do not use the reserved case as a paired lab task.

### Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

### Address these additional misconceptions

### Match the support to the first error

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MAIN LESSON · SECTION 5

## Assess the selected new case

> Keep this key until after the first attempt; accept valid alternatives with defensible reasoning.

### Selected independent question and key

Selected case: Five validation tasks have pass results A:1,1,1,0,0 and B:0,1,0,1,1. Compare performance and explain what is required before choosing a prompt.

Both pass3/5=**60%**. A alone passes tasks1 and3; B alone passes4 and5; both pass2. Examine task coverage and failure consequences using a declared **rubric**. Neither is automatically better.

Freeze the choice before testing on fresh cases.

## Judge explanation, trace and limits

| Dimension | Evidence required |
| --- | --- |
| Explanation | Names the applicable idea or procedure and ties it to the supplied facts. |
| Trace and result | Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution. |
| Limits | States the assumptions, missing evidence or conditions under which the result would change. |

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

## Give one usable repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

### W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: A prompt experiment compares complete specifications using the same task cases. Define success before inspecting **output**s; change one intended instruction and preserve other settings.

Evaluate both prompts on **paired case**s with fixed model/version, source records, **output** contract and scoring **rubric**. Use development results for selection, then reserve new final cases.

**Score** each case rather than counting attractive sentences.

## W2 Trace the worked case

On six development cases P passes cases 1, 2, 3, 4; Q passes 1, 2, 3, 5, 6. Missing-approval cases are 4 and 6. Compare totals and condition results. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: P passes 4/6 and Q 5/6. P passes one of two missing-approval cases, Q also one of two. Q improves the aggregate while leaving this condition unchanged.

## W3 Audit the limit

A report omits this qualification: Prompt selection on development cases cannot be reported as independent final performance. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: Prompt selection on development cases cannot be reported as independent final performance.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

## W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. Prompt selection on development cases cannot be reported as independent final performance.

The new case should test this particular **mechanism** rather than merely restate the original result.

## ▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

### Worked case answer

A has 4/5 versus B's 3/5 on this set. Appearance is separate. Broaden cases and repeat stochastic trials before generalising.

### Additional transfer practice question

Prompt A passes 8 of 10 tasks; B passes 9 of 10 but fails the only source-checking task. A source check is mandatory. Which can be accepted as-is?

### Reasoned practice answer

B fails a required gate despite its higher total. Inspect A's failures against all mandatory criteria; a total alone cannot establish acceptance.

### What counts as evidence

| Criterion | 0 points | 1 point | 2 points |
|---|---|---|---|
| Result | Incorrect or absent | Partly correct | Correct with necessary conditions |
| Reasoning | No supporting trace | Some steps | Clear mechanism, calculation or source support |
| Limit | Overclaims certainty | Vague limitation | Relevant uncertainty or missing fact |
| Transfer | Copies without adapting | Adapts with prompts | Solves a changed case independently |

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

### Feedback

Ask the learner to find the **evidence** that challenges this claim: “A higher overall **score** cancels a mandatory safety or **evidence** failure.” Keep the first attempt and an uncoached retry separate.

### Original transfer answer

Both **score** 3/4, but fail different cases. Inspect mandatory requirements, failure consequences and repeated-run variation.

Choose using predefined criteria, then evaluate on fresh cases that did not influence selection; do not select only favourable examples.

### Explanations for the additional workbook examples

### W1 Explain the mechanism

### W2 Trace the worked case

### W3 Audit the limit

### W4 Apply a new case independently

On eight **paired case**s A passes 1–5; B passes 1–4, 6, 7. Special cases are 5 and 8. Compare overall and special-case results; select by total passes.

Teacher explanation: A=5/8, B=6/8, so select B on development. Special-case passes A=1/2, B=0/2; the aggregate selection hides a **regression** requiring review.

### W5 Design a revealing follow-up

### Assess independent transfer

### Selected case answer

### Feedback that the learner uses

# Connect, adapt and return

> Retain the project evidence and revisit the reasoning on a changed case.

## Project evidence to keep

Evaluate Study Buddy prompts on factual, ambiguous, missing-source, and adversarial questions.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Use Study Buddy case matrices with explicit priorities, source support gates and fresh final questions.

## Support while preserving the reasoning goal

Provide one short approved passage; separate instruction, **input**, **output** and **evidence**. Stretch: introduce an unsupported claim or conflicting source and validate claim by claim with a fallback.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Selected independent record: A and B each pass3/5=60%; shared pass2,A-only1and3,B-only4and5. No task meanings, failure consequences or mandatory priorities are supplied, so neither is automatically better. State needed criteria and freeze a selected prompt before fresh final cases. Fresh retry: A=[1,0,1,0],B=[0,1,1,0]. Both2/4;shared3,A-only1,B-only2,bothfail4. Require a declared priority or further development evidence rather than an invented winner. The older eight-case exercise remains additional practice.

## Plan the delayed check

Suggested return points in this level: ai-3/10, ai-3/12, ai-3/15. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

| Record | Teacher evidence |
| --- | --- |
| Learner code and date | |
| Concept and case used; prior exposure | |
| Explanation / trace / limits | |
| Support and first repair | |
| Changed case and delayed outcome | |
| Next teaching action | |

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

## Bridge to next week

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 10: Image Generation Models. Retrieve today's **evidence** and identify what carries forward and what is new.

Next is Week10: Image Generation Models. Carry forward fixed evaluation criteria, source checks and limits on what an attractive output establishes.

### ▼ Earlier wording, alternatives and picture transcripts

Connect and extend learning

Verified Study Buddy

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case

# Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference: alternative planning checklist**

REFERENCE · SECTION 7

## Reference: alternative planning checklist

> Use as a reminder alongside the main route rather than teaching the cycle twice.

### Preparation and schedule

Teaching target: Explain the **mechanism** and terminology of controlled prompt experiments

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

## Practice and assessment reminders

Before class, solve the supported practice: On six development cases P passes cases 1, 2, 3, 4; Q passes 1, 2, 3, 5, 6. Missing-approval cases are 4 and 6. Compare totals and condition results.

Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: P passes 4/6 and Q 5/6. P passes one of two missing-approval cases, Q also one of two. Q improves the aggregate while leaving this condition unchanged.

Check every learner before discussion: Use the worked case for Controlled prompt experiments. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Prompt selection on development cases cannot be reported as independent final performance.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Remove the gate and retain total passes as the only criterion. Require a **prediction** and a reason before comparing.

Expected result: The record is a tie; another predeclared tie policy or further development **evidence** is needed. Reason: Both prompts have four passes on six questions.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

| Criterion | Not yet | With support | Independent |
|---|---|---|---|
| Explanation | Names an answer without a deciding fact | Uses a hint to link fact and decision | Explains the deciding fact in own words |
| Trace or test | Cannot connect input and result | Completes steps with prompts | Traces or tests a changed case accurately |
| Limits and revision | Claims more than evidence supports | Names a limit after a prompt | States a defensible limit and useful next test |

Project checkpoint: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference: feedback and records**

REFERENCE · SECTION 8

# Reference: feedback and records

> Use digital tools where they help retain an explanation and its evidence.

## Keep practice and independent evidence separate

Use the two formative questions as supported practice, not as independent mastery **score**s. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

### Repair and portfolio records

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.
</details>

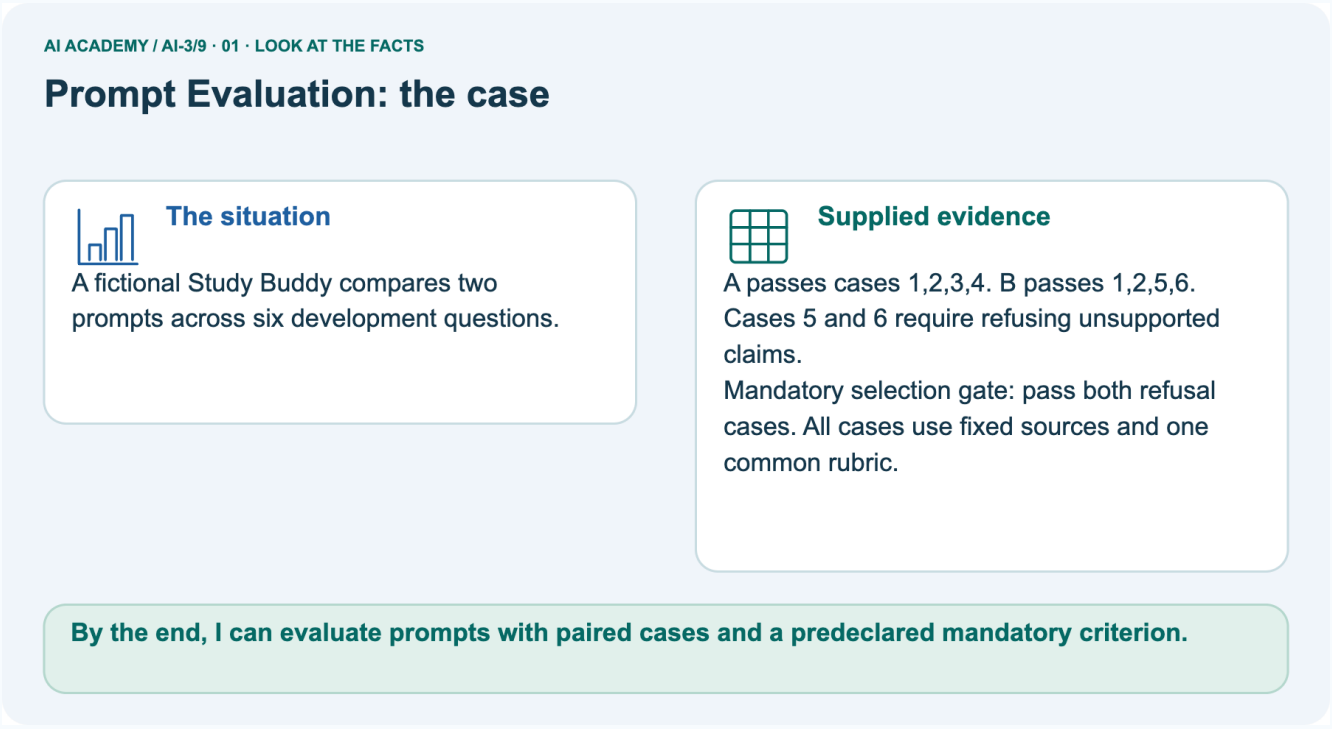
<details>
<summary>Reference: teach with the six pictures</summary>
</details>

REFERENCE · SECTION 9

## Reference: teach with the six pictures

> Choose the picture for the teaching move and connect it to the explanation.

## 01. Prompt Evaluation: the case




Follow the labelled picture with the matching explanation. Open the vector diagram

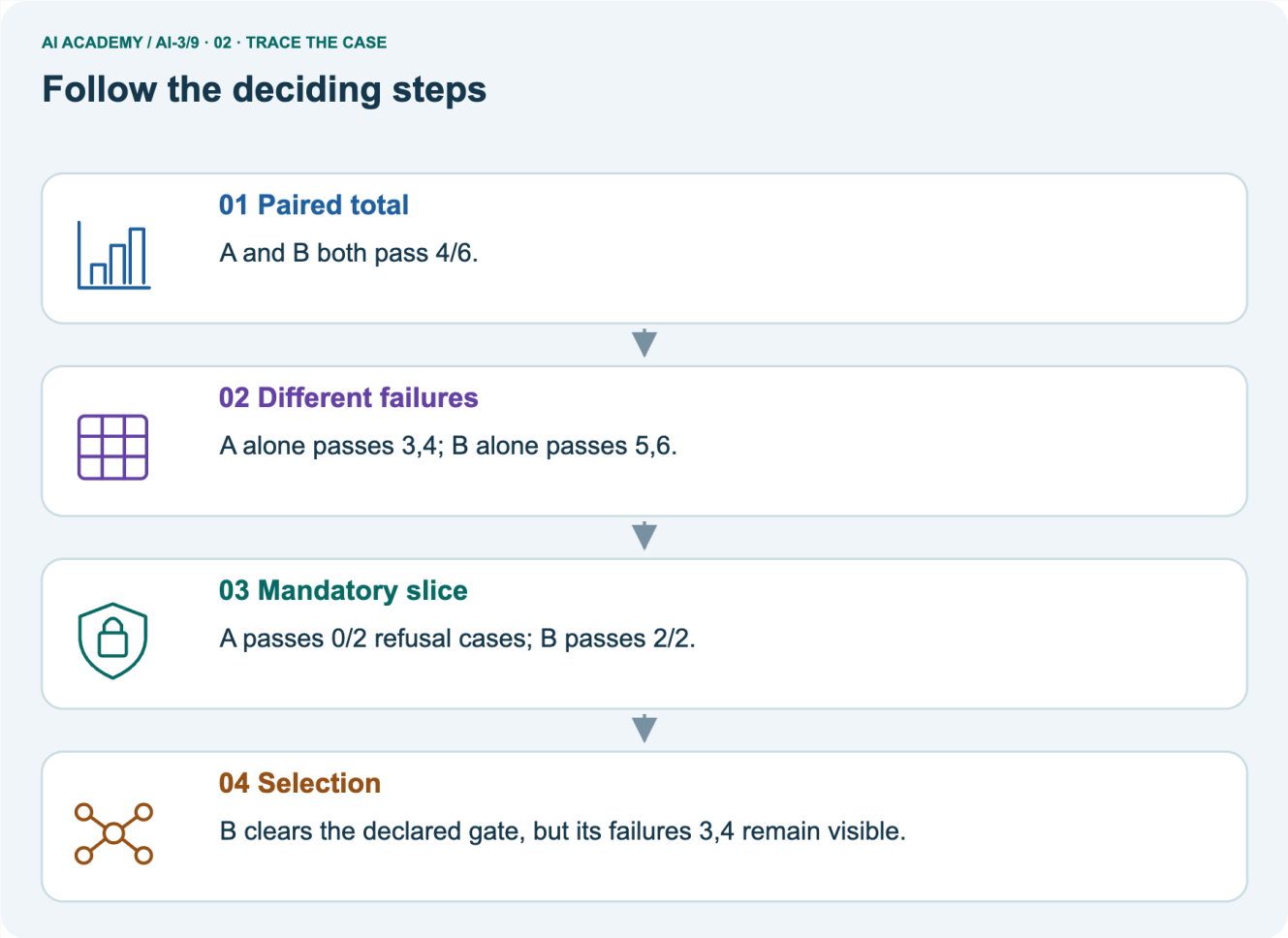
A fictional Study Buddy compares two prompts across six development questions. Supplied facts: A passes cases 1,2,3,4. B passes 1,2,5,6. Cases 5 and 6 require refusing unsupported claims.

Mandatory selection gate: pass both refusal cases. All cases use fixed sources and one common **rubric**.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

## 02. Follow the deciding steps



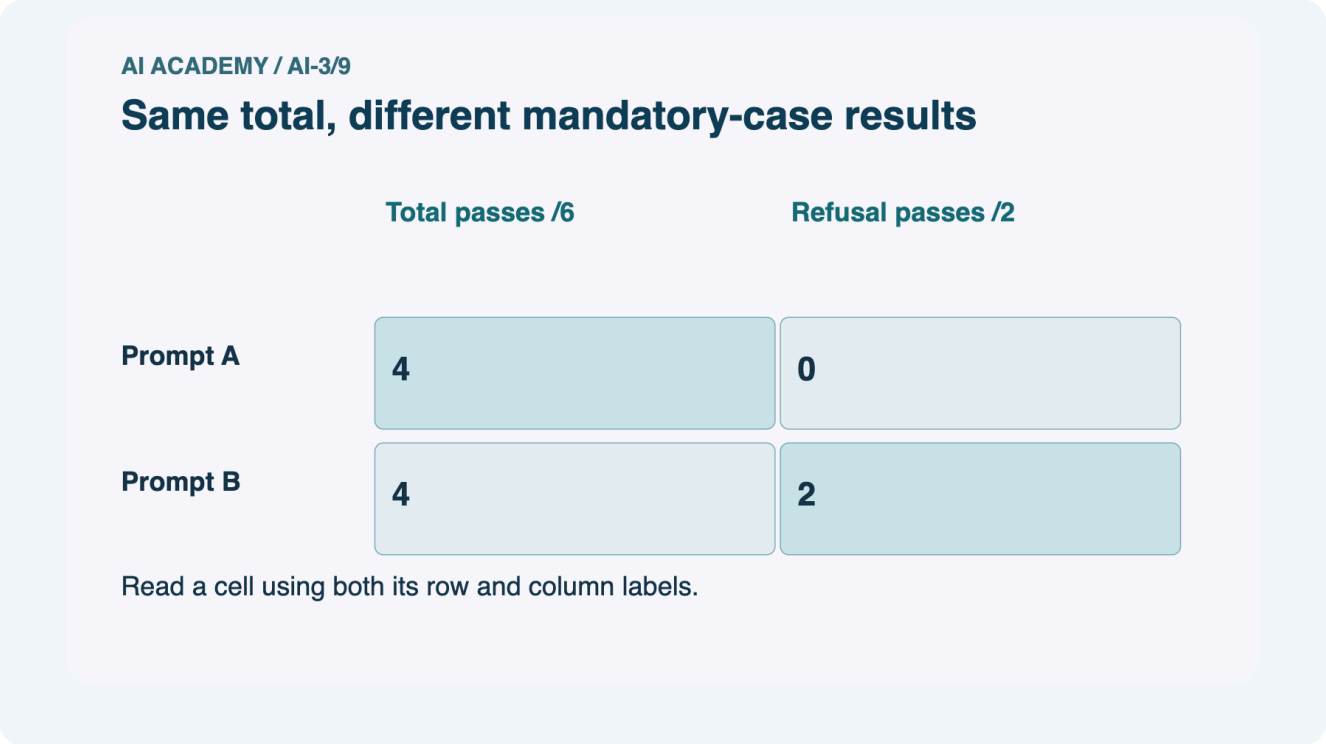

Follow the labelled picture with the matching explanation. Open the vector diagram

Paired total: A and B both pass 4/6. → Different failures: A alone passes 3,4; B alone passes 5,6. → Mandatory slice: A passes 0/2 refusal cases; B passes 2/2. → Selection: B clears the declared gate, but its failures 3,4 remain visible.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

## 03. Same total, different mandatory-case results




Follow the labelled picture with the matching explanation. Open the vector diagram

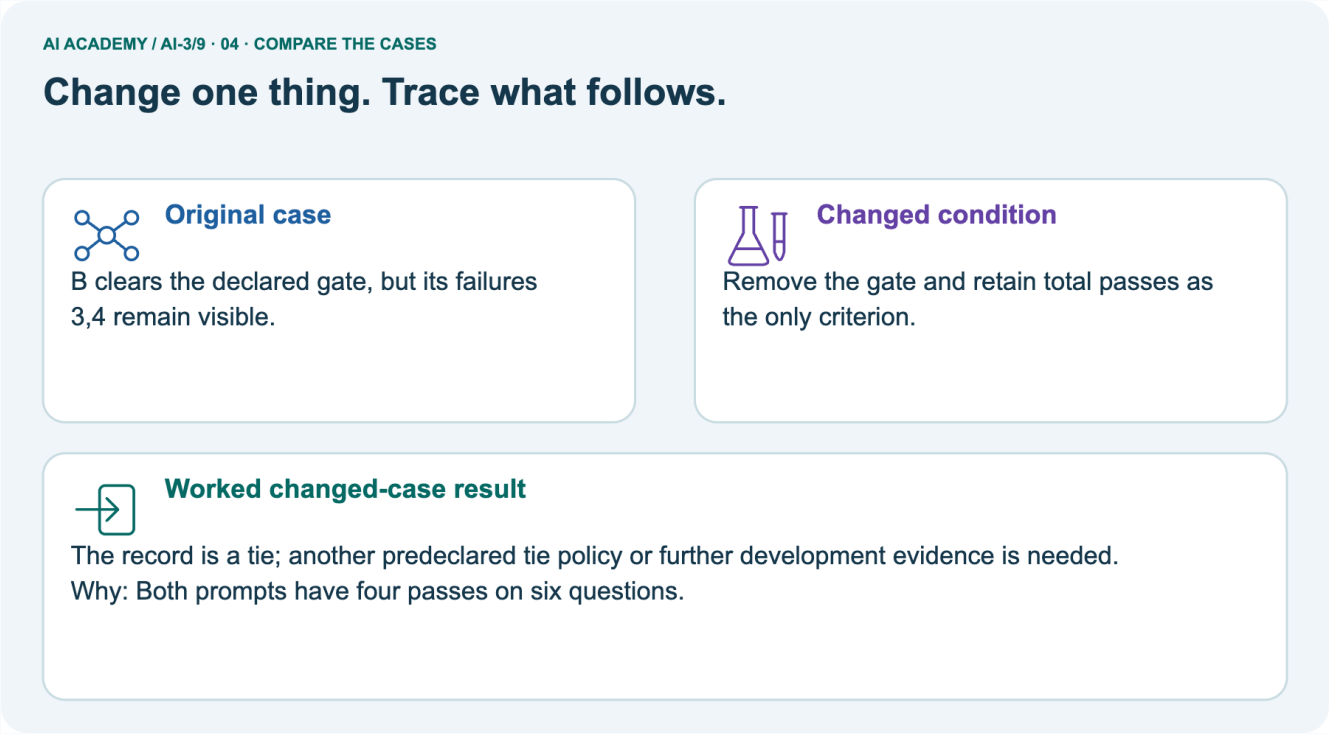
Same total, different mandatory-case results. Prompt A: Total passes /6 = 4, Refusal passes /2 = 0. Prompt B: Total passes /6 = 4, Refusal passes /2 = 2.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

## 04. Change one thing. Trace what follows.




Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: B clears the declared gate, but its failures 3,4 remain visible. Changed condition: Remove the gate and retain total passes as the only criterion.

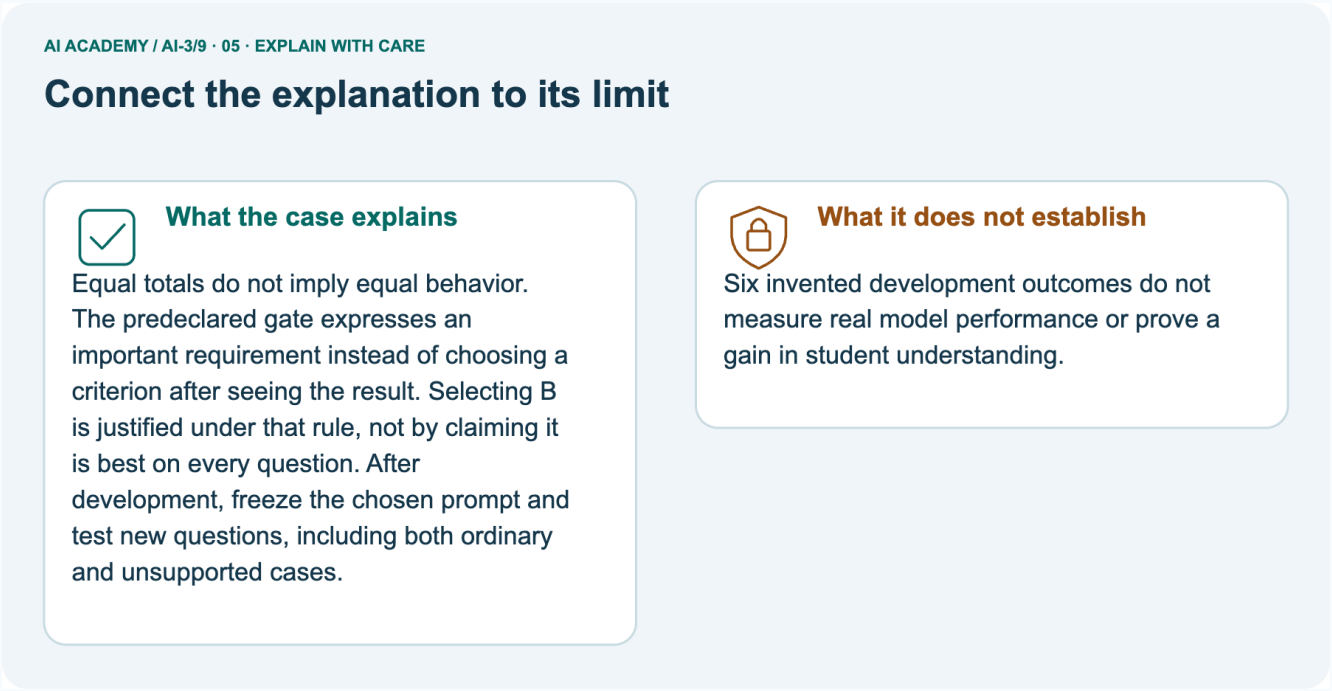
Expected result: The record is a tie; another predeclared tie policy or further development **evidence** is needed. Reason: Both prompts have four passes on six questions.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Both prompts have four passes on six questions.

## 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

Equal totals do not imply equal behavior. The predeclared gate expresses an important requirement instead of choosing a criterion after seeing the result.

Selecting B is justified under that rule, not by claiming it is best on every question. After development, freeze the chosen prompt and test new questions, including both ordinary and unsupported cases.

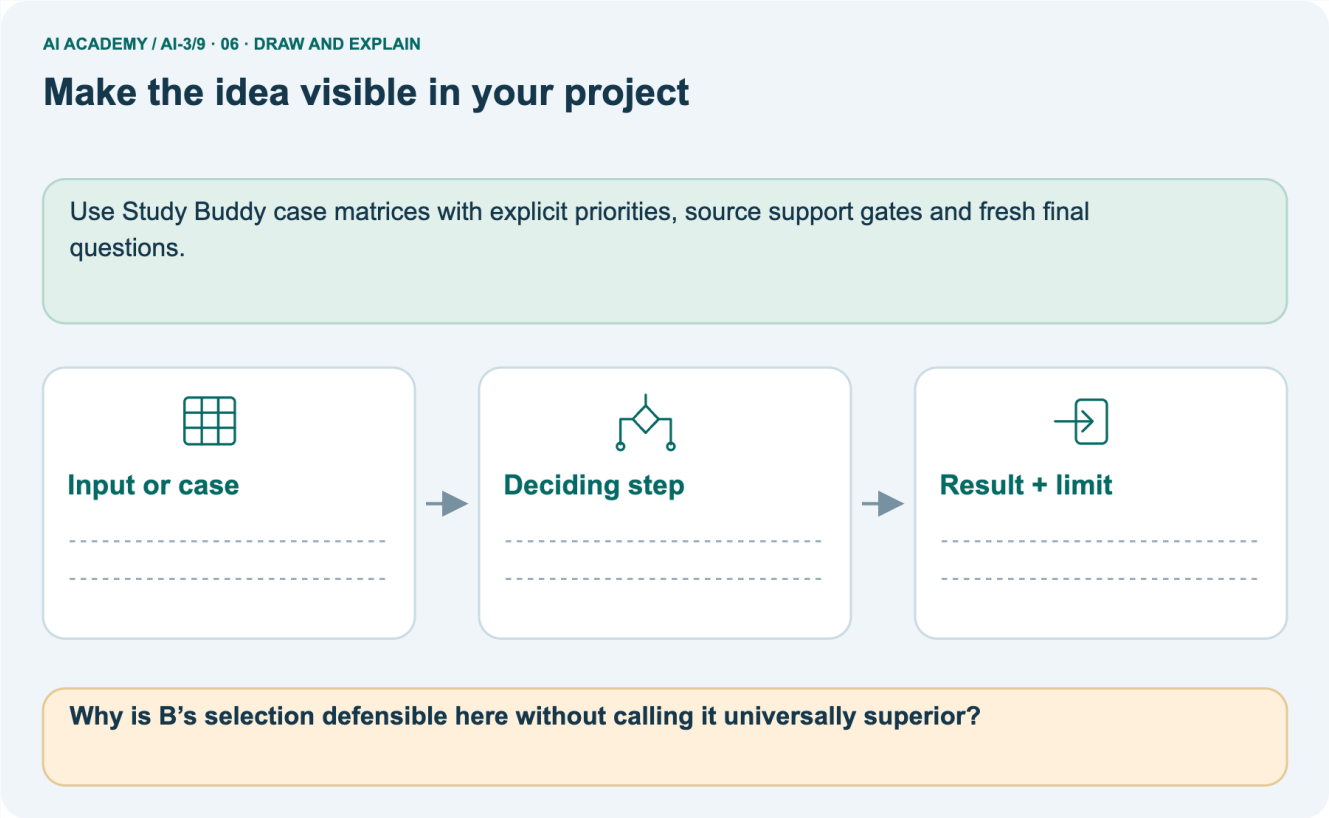
Limit: Six invented development outcomes do not measure real model performance or prove a gain in student understanding.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Six invented development outcomes do not measure real model performance or prove a gain in student understanding.

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

Use Study Buddy case matrices with explicit priorities, source support gates and fresh final questions. Draw your **input**, deciding step, result and limit.

Supported question: Why is B's selection defensible here without calling it universally superior?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: It alone passes both mandatory refusal cases. A is better on cases 3 and 4, which must still be reported.

## ▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

### 01. Prompt Evaluation: the case

### 02. Follow the deciding steps

### 03. Same total, different mandatory-case results

### 04. Change one thing. Trace what follows.

### 05. Connect the explanation to its limit

### 06. Make the idea visible in your project

## ▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.